

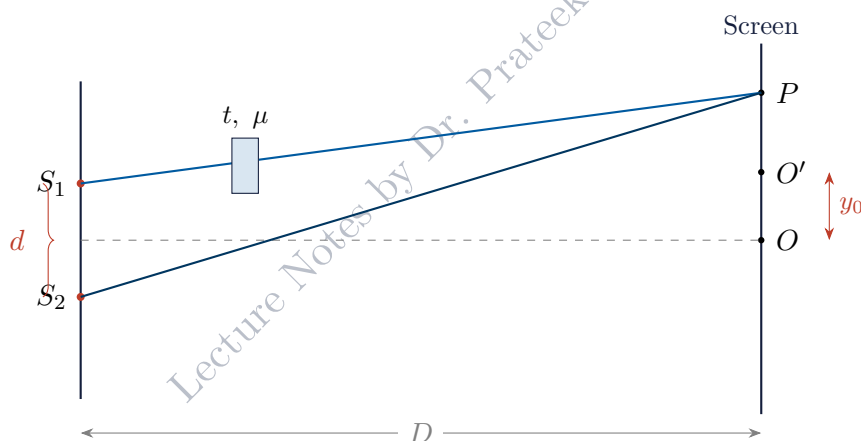
Optical Path Retardation and Lateral Shift of Fringes

B.Sc. (V Semester) — Physical Optics · Kumaon University
Unit II: Interference · Lecture 3 · Instructor **Dr. Prateek Sharma**

Effect of a thin transparent plate · Lateral shift · Fringe width unchanged · Thickness and refractive index of a thin plate

1 Optical Path Retardation

If a thin transparent plate (or film) is placed between the source and the screen, the optical path of the light passing through it is increased. This is known as **optical path retardation**. Consequently, the fringe pattern is shifted towards the beam whose optical path has been increased by the plate. The shift of the entire fringe pattern is known as the **lateral shift**.



(O : central bright fringe without the plate; O' : central bright fringe with the plate in the path of S_1 .
The lateral shift is $y_0 = OO'$.)

2 Lateral Shift of Fringes

Let S_1 and S_2 be two coherent sources of light separated by a distance d , and let O be the position of the central bright fringe. Let a glass plate of thickness t and refractive index μ be placed in the path of the beam from S_1 . Due to this, the entire fringe pattern will shift upward.

Let the central fringe shift to O' ; at the new position the two optical paths are equal, $[S_1O'] = [S_2O']$.

In travelling the distance t through the plate, the light from S_1 covers an optical path μt instead of t . Hence the path difference between the waves reaching a point P is

$$\begin{aligned}\Delta &= S_2P - [(S_1P - t) + \mu t] \\ &= S_2P - S_1P + t - \mu t \\ &= S_2P - S_1P - (\mu - 1)t.\end{aligned}\tag{1}$$

We know from Lecture 2 that

$$S_2P - S_1P = \frac{yd}{D}.$$

From Eq. (1),

$$\Delta = \frac{yd}{D} - (\mu - 1)t.$$

2.1 Position of the bright fringes

The condition for the n th bright fringe is $\Delta = n\lambda$:

$$\begin{aligned} \frac{y_n d}{D} - (\mu - 1)t &= n\lambda \\ \frac{y_n d}{D} &= n\lambda + (\mu - 1)t \\ \Rightarrow y_n &= \frac{D}{d} [n\lambda + (\mu - 1)t] \end{aligned}$$

At $t = 0$ (absence of the glass plate), the distance of the n th bright fringe from O is

$$y'_n = \frac{nD\lambda}{d}.$$

2.2 Lateral shift

The lateral shift of the n th bright fringe is

$$\begin{aligned} y_0 &= y_n - y'_n \\ &= \frac{D}{d} [n\lambda + (\mu - 1)t] - \frac{D}{d} n\lambda \end{aligned}$$

$$y_0 = \frac{D}{d} (\mu - 1)t$$

This expression is independent of n , so the *entire* fringe system is displaced through the same distance $\frac{D}{d}(\mu - 1)t$ towards the side on which the plate is placed.

3 Fringe Width Remains Unchanged

$$\begin{aligned} y_n &= \frac{D}{d} [n\lambda + (\mu - 1)t] \\ y_{n+1} &= \frac{D}{d} [(n + 1)\lambda + (\mu - 1)t] \end{aligned}$$

$\therefore$ Fringe width

$$\beta = y_{n+1} - y_n = \frac{D\lambda}{d}$$

This is the same as without the plate: the pattern is shifted, not stretched.

4 Thickness or Refractive Index of a Thin Plate

Using $\beta = D\lambda/d$, i.e. $D/d = \beta/\lambda$,

$$y_0 = \frac{D}{d}(\mu - 1)t$$

$$y_0 = \frac{\beta}{\lambda}(\mu - 1)t$$

$$\Rightarrow \boxed{t = \frac{y_0 \lambda}{\beta(\mu - 1)}} \quad \text{or} \quad \boxed{\mu = 1 + \frac{y_0 \lambda}{\beta t}}$$

Thus, by measuring the lateral shift y_0 and the fringe width β , the thickness of the plate (if μ is known) or its refractive index (if t is known) can be determined.

Lecture Notes by Dr. Prateek Sharma